Methodological Integration:

Lessons from the Ecoregional Initiative

for the Humid and Sub-Humid Tropics of Asia

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Empowerment of stakeholders

Partnership hierarchies

Abstract

Integrated natural resources management (INRM) for sustainable agricultural development has to address both the livelihood goals of farmers and ecological sustainability of agro-ecosystems and the natural resource base. New tools need to be developed to take conventional methods of research and analysis further by taking various spatial scale levels and their interactions into account, quantifying trade-offs between bio-physical sustainability and socio-economic considerations at different scale levels, and quantifying trade-offs between objectives that different users of the land and natural resources may pursue. Under Ecoregional Initiative for the Humid and Sub-humid Tropics of Asia - Ecor(I)Asia - one major set of activities has been the development of approaches and tools to meet the challenges of INRM research for sustainable agricultural production. Examples provided illustrate the role of these methodologies in the three main phases of knowledge development for improving INRM impact, i.e. knowledge generation, knowledge capitalization and knowledge mobilization. These methodologies and are designed for better integration across disciplines and across hierarchical scales of organization. The case is made in this paper for moving towards methodology integration, recognizing that no single set of

methodologies can solve complex resource use planning and INRM problems, and that methodologies integrated in a complementary manner can help bridge the top-down and bottom-up approaches in INRM. Inherent in the development and implementation of these methodologies is the forging of partnerships and in fostering linkages with multiple stakeholders as well as using the knowledge base and integrative tools as communication platforms.

Introduction

Research challenges in natural resources management

As the world's growing population place greater demands on the goods and services of its finite natural resources, there are two main challenges faced in natural resources management (NRM) for sustainable agricultural production (taken loosely to include crop, livestock, agro-forestry production). The first challenge is to seek viable alternatives of production systems that can meet the livelihood objectives of farmers at the same time sustain the natural resource base on which they depend. The second challenge is to apply location-specific findings from NRM research and development efforts to wider domains, for two important reasons. One reason is that concerns of ecological sustainability cover broader geographical areas than plots, fields and farms. The second is that NRM research ought to be able to benefit large numbers of people, particularly the poor, more extensively.

In having to deal with increasingly complex situations of multiple uses and users of the resource base at multiple scales, NRM must be based on an understanding of the existing interactions between the resource base and social dynamics, as well as the relationships among several levels of organization. There is a felt need for research and development (R&D) efforts in NRM to adopt more "integrated" approaches. There are many facets of integration, which give rise to different perceptions of what integrated natural resources management (INRM) entails.

Dimensions of integration

In order two meet the two challenges of effective NRM mentioned above, the basic precepts of integration for NRM would be to (i) go beyond sectoral considerations to an inter-disciplinary perspective, (ii) go beyond site specificity, and (ii) strengthen linkages along the research-development-policy continuum (Figure 1).

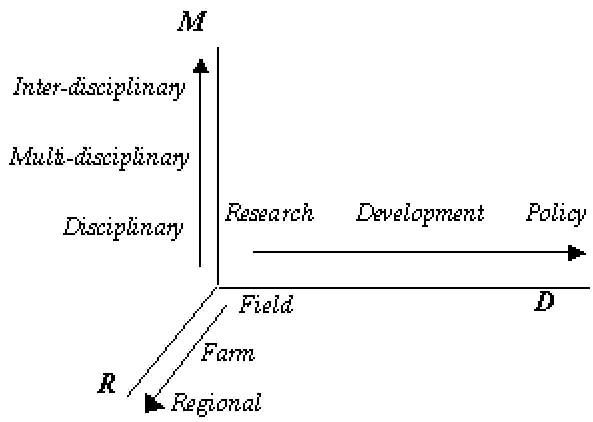

Figure 1. The dimensions of integration in natural resources management

An inter-disciplinary approach to NRM is important because while the effects of natural resources over-exploitation are more commonly studied as bio-physical phenomena, the underlying causes are largely socio- economic and institutional/political. The second dimension relates to scale issues in NRM. One aspect of scale is the cumulative effects of individual/community uses of the natural resources base, as the different ecological processes that underpin the state and flow of natural resources operate at different scales. Another aspect is that socio-economic and institutional/political forces, and the interests and concerns of different stakeholders, are also directed at different scales. While socio-economic gains from using the natural resources are most immediate and evident at the household/community level, the costs to the environment and the resource base are not borne by the same groups who benefit, or even the same generation. The third dimension is important because NRM research findings can only have broad impacts if the research is closely linked with development efforts, and the enabling social, institutional and organizational conditions for effective INRM are nurtured. Furthermore the success of R&D efforts in INRM hinges upon effective policy and governance instruments that are supportive of the diverse needs of a variety of users yet protective of the long-term productive capacity of the natural resources.

The Ecor(I)Asia

This paper relates the experience of Ecoregional Initiative for the Humid and Sub-humid Tropics of Asia[1] - Ecor(I)Asia – in developing a number of ecoregional approaches (Rabbinge, 1995), methodologies and tools to address issues of sustainability in NRM for agricultural production. Although varied, these methodologies are designed for better integration across disciplines and across hierarchical scales of organization to achieve a better understanding of the interactions between the natural resources and social dynamics. In developing these methodologies, attempts were also made to involve stakeholders to enhance relevance as well as a sense of ownership, acceptability and utility of these tools.

[1] One of the eight ecoregional programmes of the Consultative Group of International Agricultural Research Centers (CGIAR).

Demands on INRM

A case example of an NGO-government collaborative agricultural development effort in the Red River Basin of Vietnam illustrates the basic challenges of attaining food security, and ecological and economic sustainability in the agricultural production activities of a number of communes in Tam Dao district of Phu Tho province. The project adopts a highly participatory approach in identifying problems and key intervention points to break the spiral of non-sustainability (Figure 2) that existed and to turn the situation around to one of a more sustainable nature (Figure 3). The salient features of the project are given in Appendix 1.

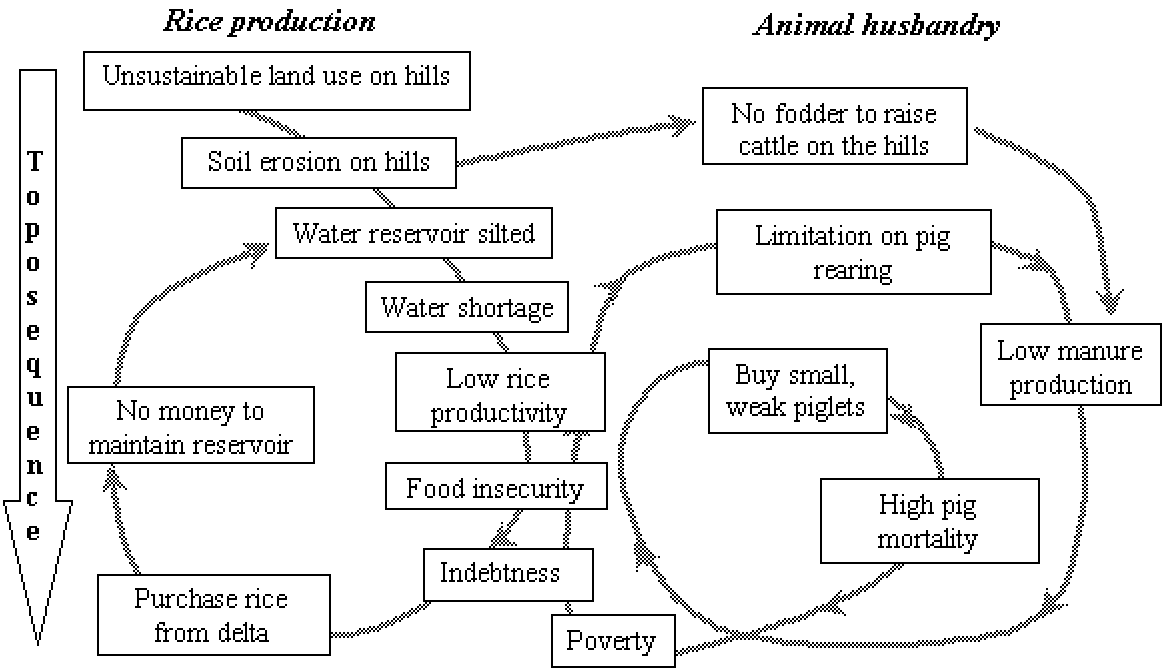

Figure 2. The spiral of non-sustainability: Case example of Tam Dao, Vietnam.

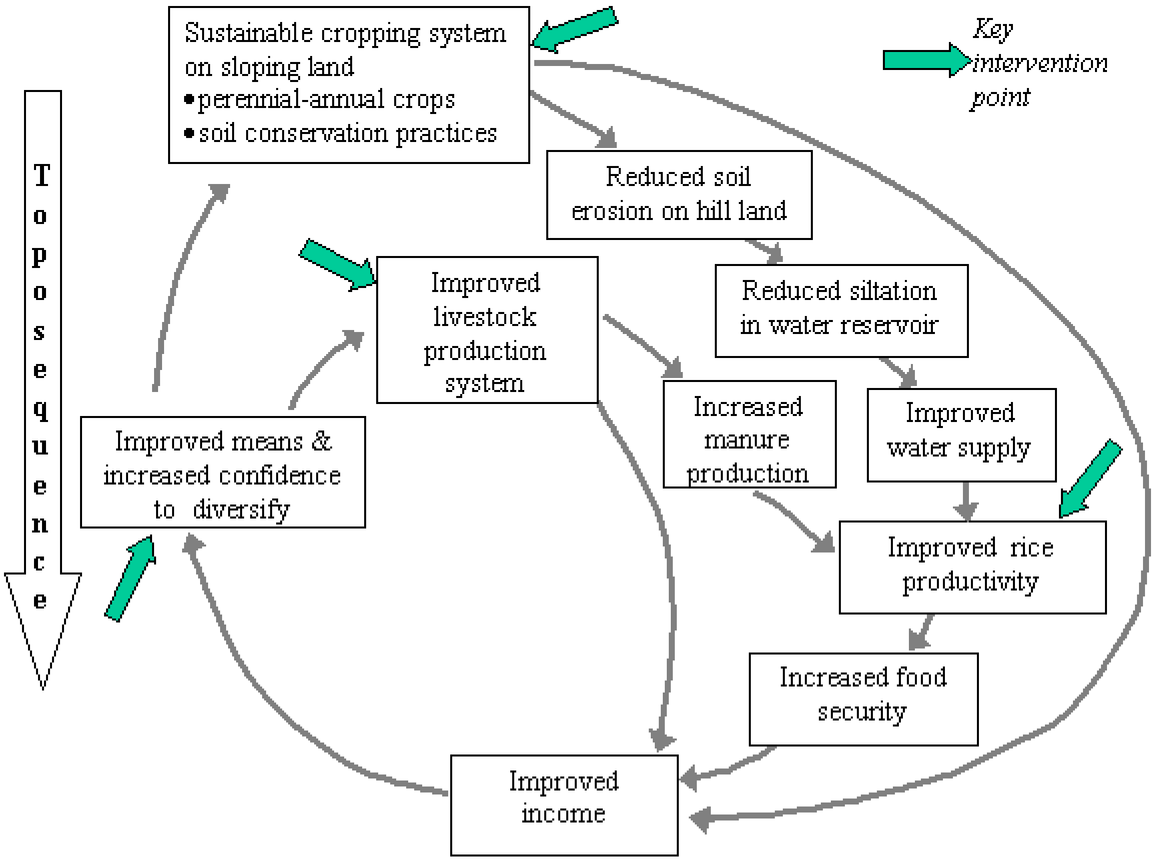

Figure 3. The cycle of sustainability: Case example of Tam Dao, Vietnam.

Several such success stories have emerged from the 10-year efforts and experience of the Red River Program. Although spontaneous diffusion of certain types of innovations to neighboring villages does occur, the project teams recognize their limitations in being able to "multiply" their successes to large areas in an effective and timely manner, other than to repeat the same time- and resource-intensive efforts at each location in turn. This highlights the need for forging strategic partnerships and developing closer research - development linkages to attain wider impact of INRM. Research support is needed for many aspects:

a. in providing technical and scientific input into developing and testing innovations, both indigenous and modern;

b. in systematic characterization and analysis to determine the key determinants of the success factors;

c. in determining target areas for effective delivery of INRM practices; and

d. in documenting the operational methodology and quantifying the impacts of INRM practices on livelihood improvement and conservation the natural resource base.

Such case studies provide partly the motivation and impetus for the methodological development efforts under Ecor(I)Asia, in particular to address the complex issues of NRM in a more holistic and integrated manner.

Framework for knowledge development for INRM

We maintain that sustainable NRM lies ultimately in the hands of the stakeholders. For stakeholders to make informed decisions on how they manage their resources they need to have the knowledge and understanding of the key NRM issues. The multiplicity of stakeholders (farmers, communities, local government administrators and policy makers), with different resource use objectives and strategies, underlines the need for skills and tools to facilitate communication and negotiation in managing common pools of land and water resources.

The fundamental role of INRM research is knowledge development for empowering people to manage the natural resources on which they depend. We identify three main phases of knowledge development, i.e. knowledge generation, knowledge capitalization and knowledge mobilization, not only as being imperative for effective INRM, but are equally valid for the transfer of specific technologies such as crop, soil, water, pest management practices (Table 1). In fact, INRM aims at increasing the relevance and impact of well founded, discipline-based, scientific NRM research and to achieve this, more integrative approaches and tools need to be developed.

Table 1. Phases of knowledge development for INRM.

Technology development for NRM		**Phases of knowledge development**	R&D linkage for attaining impact of Integrated Natural Resources Management				
Objective	Activity		Objective	Activity	Soil erosion study, North Thailand	SAM example, Red River Basin of Vietnam	LUPAS example, Red River Basin of Vietnam
Seek promising technologies	Experimentation and characterization	**Knowledge generation**	Understand the existing situation - ecological processes - social dynamics - market	Characterization	Multi-scale diagnostic study of changes in erosion risk in relation to changes in cropping systems	Analysis of agrarian systems evolution and differentiation; compilation of innovations data base	Resource evaluation Yield estimation Input-output estimation
Develop innovation	Field testing, demonstration and distillation of scientific knowledge to simple rules	**Knowledge capitalization**	Add value and make use of knowledge	Development of tools and methodologies for decision support	Multi-agent systems (MAS) modeling of land use change dynamics and soil erosion	Testing of sustainable NRM innovations; SAMBA multi-agent system based simulation modeling	Interactive Multiple Goal Linear Programing
Deliver technology to user	Transferring of technology to users	**Knowledge mobilization**	Empower stakeholders	Transferring and implementation of decision support systems	Using MAS as "companion modeling" to support discussion and negotiation among multiple stakeholders	Establishment of resource center; Making knowledge base accessible to local community Using SAMBA as role game	Stakeholder consultation, training and operationalizing of decision support system

Notes:
SAM = Mountain Agrarian Systems

LUPAS = Land Use Planning and Analysis System

Knowledge generation required for INRM emphasizes a more holistic approach to understand the circumstances in which natural resources are being used, the needs for and implications of introducing NRM interventions in the context of ecological, socio-economic and policy environments. Tools are needed to capitalize the knowledge generated in order to address the diverse objectives of stakeholders and, recognizing that there may be non-compatible demands, to determine trade-offs between bio-physical sustainability and socio-economic considerations at different scale levels. These tools, in turn, should be used to mobilize the knowledge and empower stakeholders in their decision making and management of their natural resource base.

Case examples

Three examples (summarized in Table 1) are presented to illustrate how the framework described above is implemented under the Ecor(I)Asia.

1. North Thailand soil erosion risk study

A study was conducted in the Mae Salaep watershed in Chiang Rai province, North Thailand (Trebuil, et al., 2000) to determine the effect of diversification of farming systems in response to market integration on land degradation risk.

Knowledge generation: At the field level, an on-farm diagnostic survey analysed the influence of the main cropping systems on the risk of soil erosion under various slope and climatic conditions At the farm level, a typology of rapidly diversifying household-based farming systems was developed to understand farmers' differentiated management strategies. At the watershed level, a GIS-based analysis of land use changes was carried out for the period 1990-98.

Knowledge capitalization: Multi-agents system (MAS) modelling (Ferber, 1999) was applied to document and link the interactions between soil erosion and cropping systems dynamics, and to facilitate knowledge integration across scales and disciplines (Bousquet, et al., 1999), using data sets and information collected at the three levels of organisation, i.e. fields, farms and village/watershed, as depicted in Figure 4. Details of the MAS approach are given in Appendix 2.

Figure 4. Schematic flow diagram for watershed soil erosion modeling using multi-agent systems, North Thailand.

The study found that several practices adopted by farmers in diversified farming systems actually improved soil and water conservation (smaller fields, terraced paddies replacing upland rice fields on sloping land, and more perennial crops), and that these dynamics appear to contribute more to the overall soil conservation of the watershed than the past 15 years' efforts at introducing externally-imposed soil and water conservation techniques. MAS modeling is used to simulate and identify agro-economic conditions that are consistent with farmers' dynamics yet improve further soil conservation. Figure 5 shows mapped results depicting yearly changes in cropping systems and corresponding changes in erosion risk.

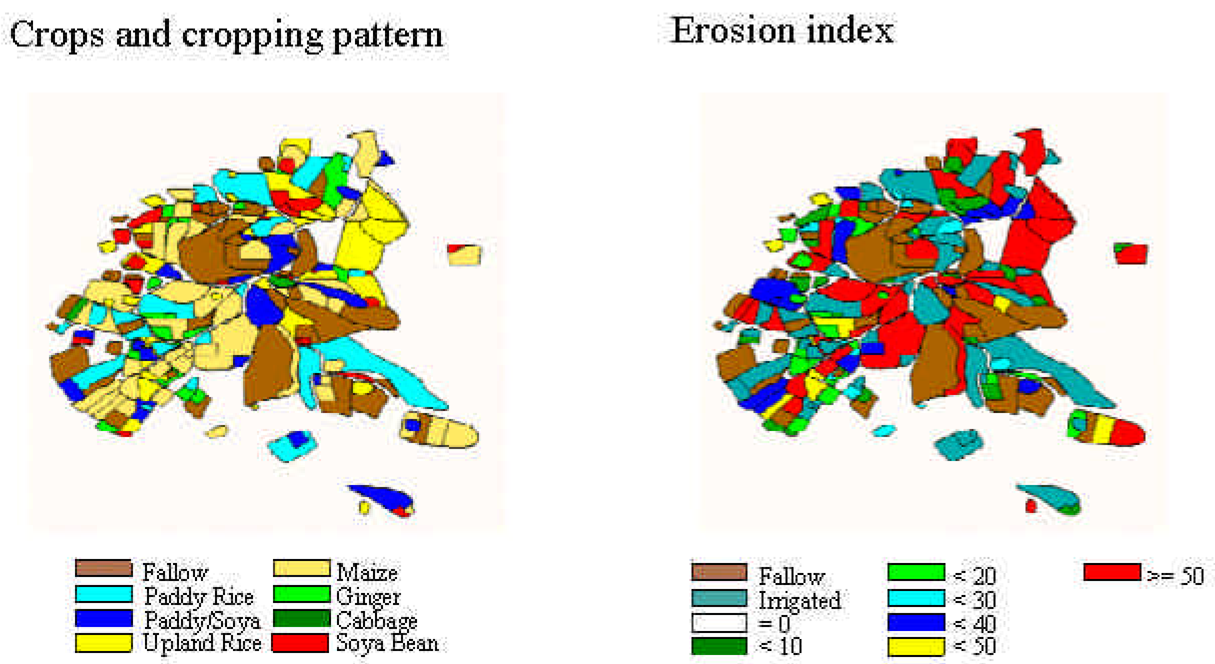

Figure 5. Example of outputs from the MAS model for watershed erosion risk analysis, Mae Salaep watershed, North Thailand.

Knowledge mobilization: In a more participatory way, a simplified version of the MAS model is being translated into a role game to be used as a decision support tool by local administration agencies and farmer communities in a negotiated mode of NRM. A local government agency will assist in field-testing the role game with farmers and sub-district administration officers at 14 sites across north Thailand.

2. Mountain Agrarian Systems Project, Red River Basin of Vietnam

In the northern uplands of Vietnam population pressure, changing political systems and policies on land allocation (from collective to individual management) resulted in shifts in resource endowments of ethnic groups, influencing their ability to respond to food production needs and their level of exploitation of the natural resource base across the toposequence (Castella, et al., 1999a). Farmers, in having to respond to these changes and intensify production, often lack the knowledge to seek alternatives to their traditional slash and burn practices which have become non-sustainable.

The Mountain Agrarian Systems (SAM) Project in the uplands of the Red River Basin aims at improving food security of the ethnic minority groups while ensuring the sustainability of the agricultural production systems and the natural resource base of the fragile environment. The Cropping Systems component of the project (SAM-CS) concentrates on seeking viable alternatives to the present unsustainable production systems at field/farm level, while the Regional component (SAM-R) focuses on developing research and operational methodologies to scale out location-specific studies.

Knowledge generation: Under the SAM-CS component, participatory testing and evaluation of technical innovations were carried out e.g. on-farm testing of upland rice varieties, direct seeding under grass and legume cover crops, mulching of upland rice and maize, and introduction of new winter crops. Encouraging results have been obtained, e.g. yield increases, increased organic matter incorporation, soil stabilization against erosion have resulted from direct seeding and mulching practices for upland rice. Knowledge generation under the SAM-R Component focuses on developing multi-scale approaches to characterize and understand the evolution of agrarian systems changes in the context of bio-physical and socio-economic and policy environments and to evaluate the performances of existing and promising cropping and farming systems, using a combination of field, survey and remote sensing techniques (Castella, et al. 2000).

Knowledge capitalization: GIS tools are used to consolidate the knowledge gathered at the various spatial and temporal levels as well as across the various disciplines. A multi-agent systems based model, SAMBA, was developed to mimic the process of farming systems differentiation over a 40-year period in order to determine the driving forces of the observed changes. Figure 6 illustrates how SAMBA is used to test if a limited number of household socio-economic characteristics can sufficiently explain the differentiation of household typology in terms of production strategies. This modeling approach allows aggregated results of individual household behavior to be represented at the village level, and likewise the scaling up from village to commune level. It provides the basis for developing a regional model, i.e. a multi-disciplinary parallel of "pedo-transfer functions", thus allowing for up-scaling of research and development findings in INRM.

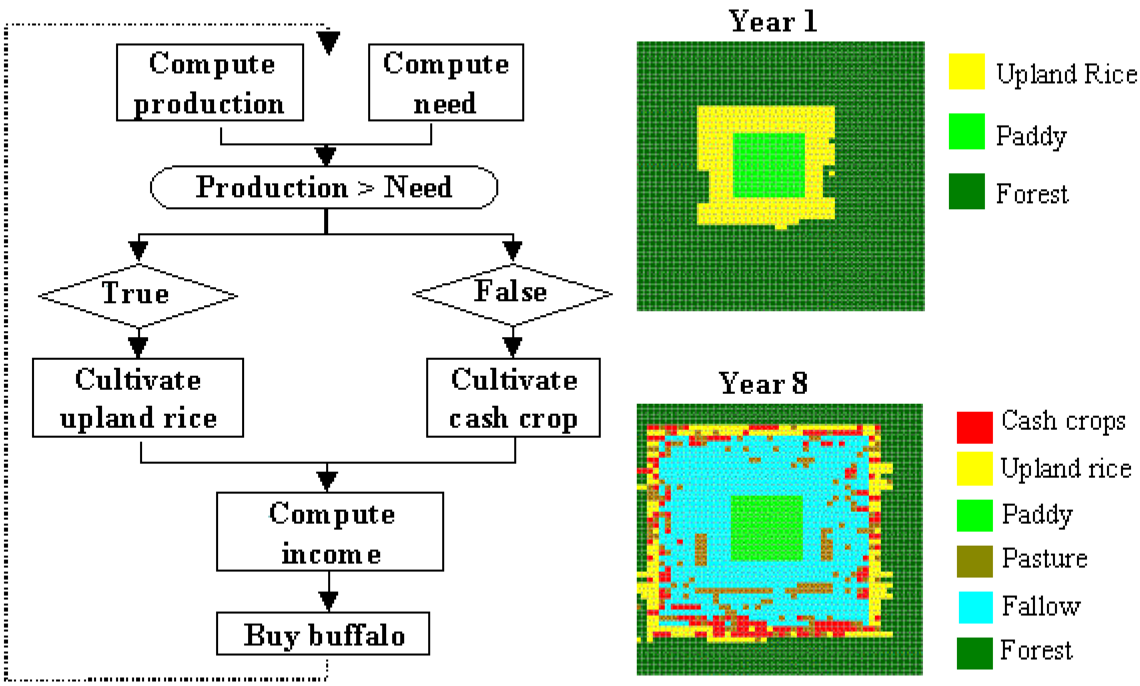

Figure 6. The SAMBA model applied to simulation of land use changes, SAM-R study in the uplands of the Red River Basin, Vietnam.

Knowledge mobilization: Replication of the technical innovations developed by SAM-CS is being carried out in the network of communes. A resource center has been established at the provincial capital, where the databases from relevant studies are deposited. The resource center provides the focal point for interactions among R&D partners and stakeholders. Converted into a role game, SAMBA is being developed as a tool to explore options and anticipate implications of current and promising NRM practices. It promotes mutual social learning, whereby the interactions that it elicits among stakeholders provides better insight into the local social dynamics.

3. Land Use Planning and Analysis System (LUPAS)

Another major methodological development under the Ecor(I)Asia addresses concerns over use of natural resources at regional level, and linkages with policy and land use planning decisions. LUPAS, a computerized decision support system (DSS) was developed, which applies interactive multiple goal linear programming approach to address conflicting land use objectives by stakeholders at different levels, by quantifying trade-offs between these objectives and between bio-physical sustainability and socio-economic considerations (Hoanh, et al., 1998). The LUPAS methodology was developed under the SysNet (Systems Research Network for Ecoregional Land Use Planning in Tropical Asia) project with four case studies in India, Malaysia, Philippines and Vietnam (Roetter and Hoanh, 1998), each with different characteristics and different NRM concerns.

The conceptual structure of LUPAS is based on three main parts (Roetter and Hoanh, 1999), as shown in Figure 7. The process of LUPAS development involves iterative interactions among research partners and stakeholders (Hoanh and Roetter, 1998), as shown in Figure 8.

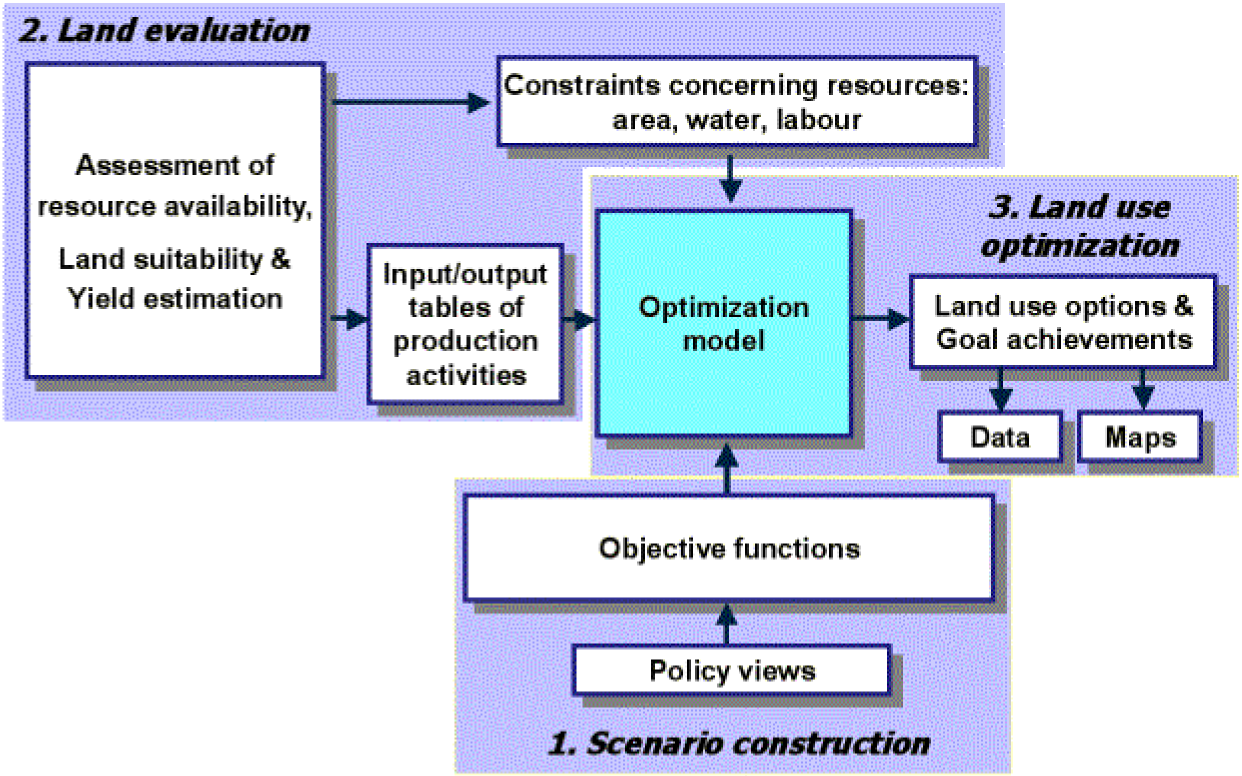

Figure 7. Conceptual model of the Land Use Planning and Analysis System (LUPAS).

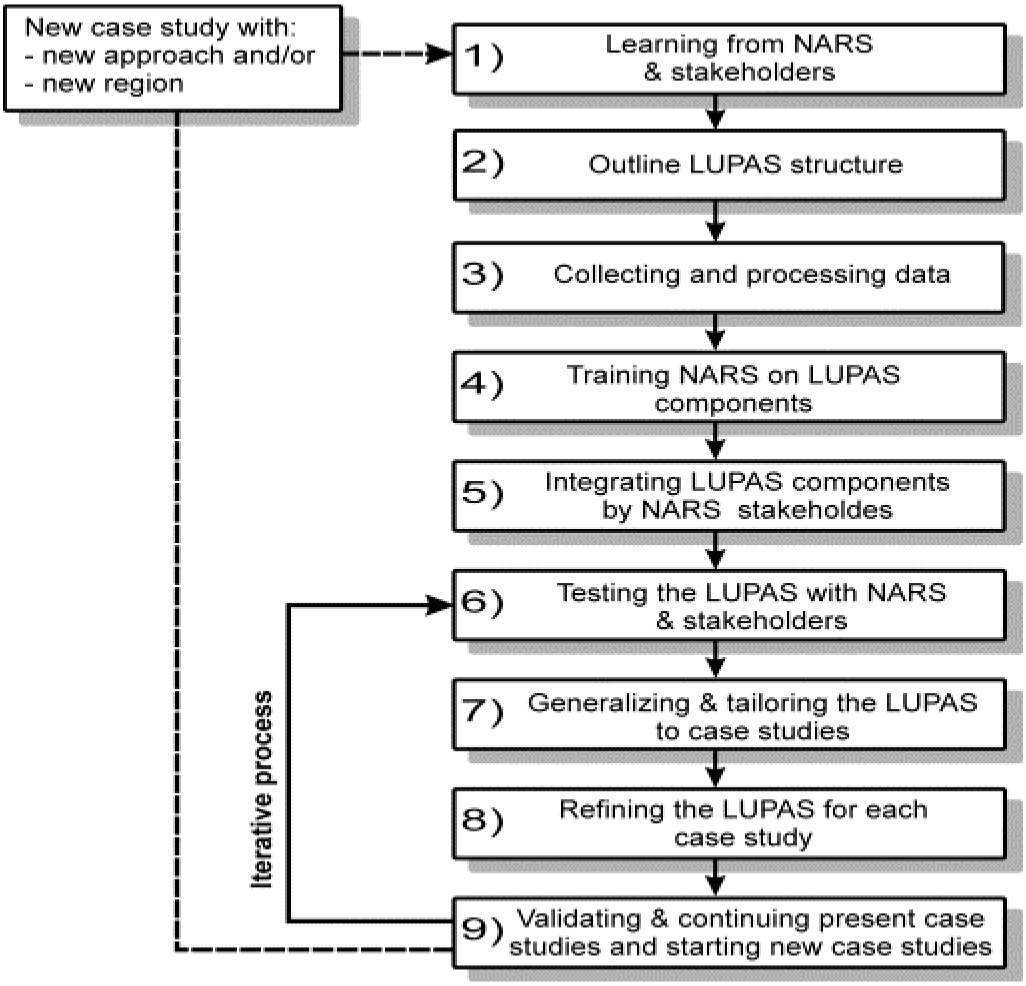

Figure 8. The sequence of LUPAS methodology development.

Knowledge generation: In Component 1 of LUPAS (Figure 7), the major characteristics of the region are determined and objective functions are formulated based on the policy views of stakeholders on the direction of development. In component 2, bio-physical and socio-economic data are collected to develop a knowledge base on the resources, on which basis the existing and promising land use types, and their requirements on both natural and human resources are analyzed.

Knowledge capitalization: The land use optimization (Component 3 in Figure 7) constitutes the core for knowledge capitalization in LUPAS. An Interactive Multiple Goal Linear Programming (IMGLP) model is developed to match the resource base with the objective functions selected by stakeholders and to generate the optimal land use options for the region, given certain bio-physical, socio-economic and policy constraints.

Knowledge mobilization: Knowledge on the resource base and the IMGLP model are refined and transferred to the NARS and stakeholders during the iteration process from step 6 to step 9 in

Figure 8. Stakeholders in the case study regions are already making use of the results to review their resources and revise their regional land use plans.

Methodology integration

The major outputs from the above three examples are tools that facilitate integrative analyses and exploration of options regarding the use and management of natural resources, as well as operational methodologies to involve stakeholders in the development and use of these tools. While the Ecor(I)Asia started with developing a “basket” of tools for integrated approaches to INRM, we are now moving towards integration of these methodologies, recognizing that no single set of methodologies can solve complex INRM problems and that methodologies integrated in a complementary manner can be more effective and have greater impact.

Figure 9 illustrates the prospect of methodology integration within the Ecor(I)Asia, particularly the LUPAS and SAM methodologies in the northern uplands province in Vietnam. LUPAS allows for exploration of development scenarios at regional level, assuming certain stakeholder objectives under existing and expected future conditions of the resource base. It provides a rational basis for policy makers to make informed choices of preferred development options, having an understanding of the trade-offs, implications and consequences of the various options. In order to move from the existing to the preferred scenario, it still remains to identify the development pathway, and to introduce interventions to steer changes along this pathway.

For INRM, the chosen development pathway ought to be based on long-term sustainability. However the longer term goals of resource conservation and environmental protection are not likely to be realized by adopting a solely top-down approach if these are not reconciled with the livelihood concerns and acceptability of interventions by the immediate users of the resource base. In this regard, the bottom-up approach adopted and methodologies and tools developed by the SAM-Regional project help strengthen community capacity to develop livelihood strategies that are more sustainable as well as to influence decisions and policy interventions that are supportive of their needs and interests. Integration of the LUPAS and SAM-R methodologies would help bridge the gap that commonly exists between top-down and bottom-up approaches to INRM.

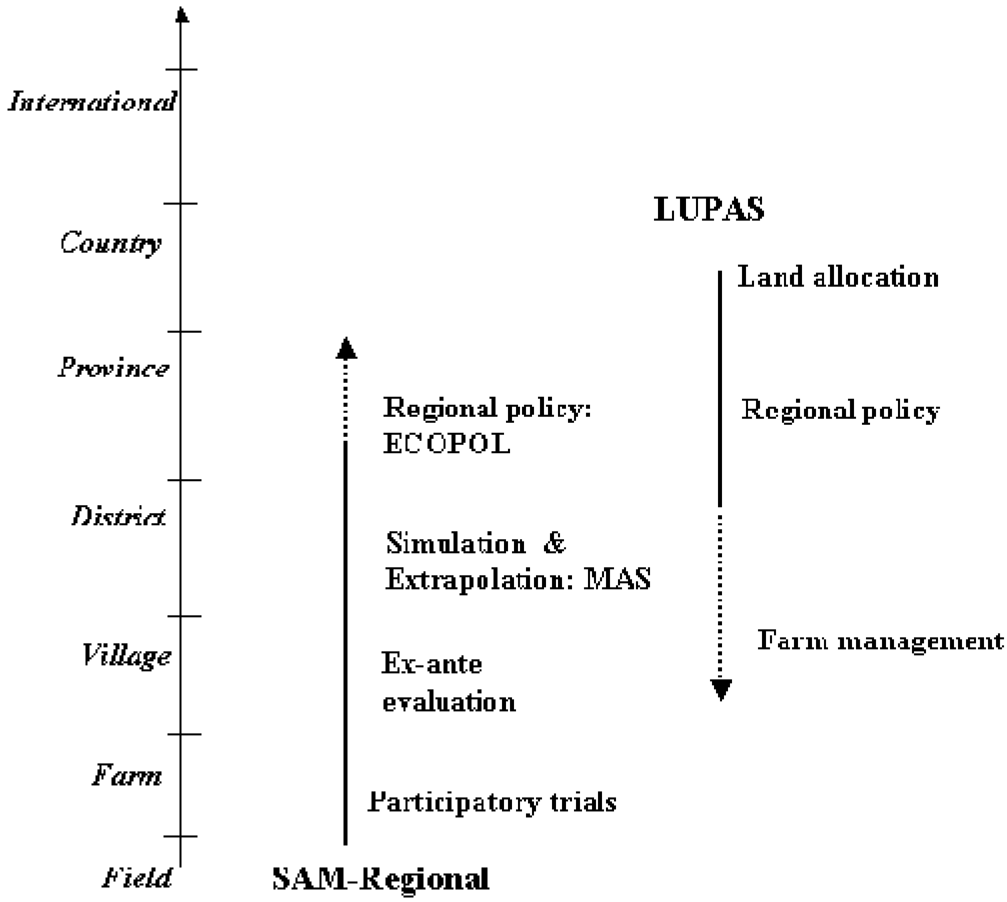

Figure 9. The dimensions of integration in natural resources management

Forging partnerships

Inherent in the development and implementation of these methodologies is the forging of strategic yet dynamic partnerships among projects and organizations, and in fostering linkages with multiple stakeholders. For example, the SAM project involves partners from international and national research organizations working closely with the communities and local government units. In turn, SAM works closely with other complementary projects under the umbrella of the host national institution (Figure 10). The introduction of LUPAS into the same geographical area adds on new partners under the coordination of the National Institute of Soils and Fertilizers (NISF). The LUPAS-SAM methodology integration is being done by an interdisciplinary and multi-institutional team of researchers, thus forming a second level network that builds linkages between institutions that used operate independently. Such a partnership network constitutes the key institutional support for tackling multi-scale, complex NRM issues in a holistic manner (Castella, et al., 1999).

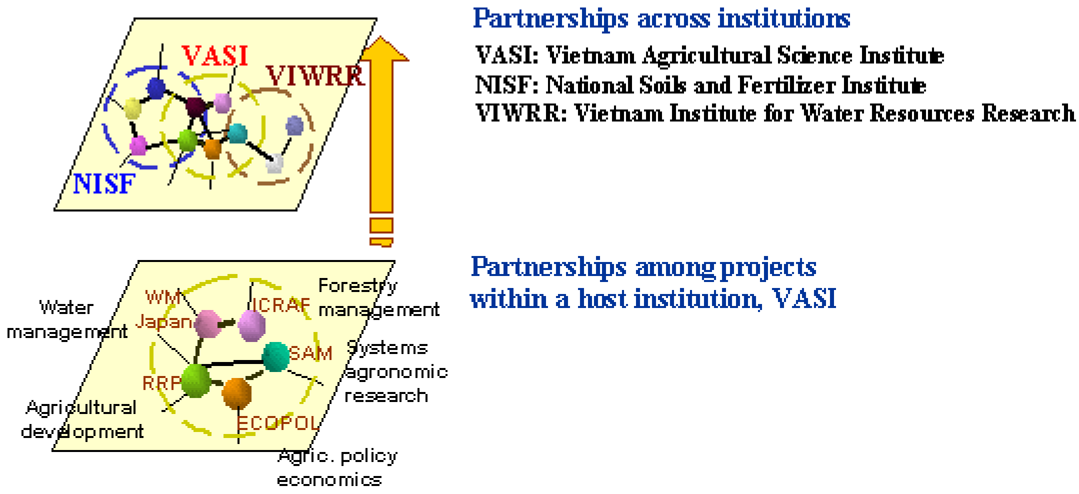

Figure 10. Building the partnership hierarchy in the Red River Basin ecoregional site, Vietnam.

Conclusions

The experience of the Ecor(I)Asia provides us with some insights into the likely success factors in developing and implementing INRM approaches for sustainable agricultural production.

1. The methodologies for developing INRM need to be highly adaptive as no single method will prove successful everywhere. While methodological diversity is needed to tackle complex NRM issues, it is also important that there is complementarity in addressing varying scale and thematic perspectives. These new integrative methodologies are not meant to replace but to augment the well-founded scientific outputs of the various disciplines, and to attenuate their impacts by bringing research closer to the source of the problems and the targets for their solution.

2. The research challenges in INRM are not only confined to developing scientific and technical solutions to managing the natural resource base, but also in devising operational methodologies that are participatory and facilitate the linkages among research, development and policy implementation. On both counts, the researcher takes on the role of a facilitator in an interactive learning process rather than an expert or lessons giver. Building the operational framework to tackle NRM issues itself constitutes a research challenge, and process documentation becomes integral to research execution and the outputs.

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Appendix 1

The Tam Dao Community-based Rural Development Project in Phu Tho Province of Vietnam

This case example is one of several rural development projects under the Red River Program, which has a network of five research and development sites in the uplands, midlands and delta of the Red River Basin of Vietnam. The Program is a collaborative effort between a French NGO, Groupe de Recherche et d'Echanges Technologiques (GRET), which provides coordination and technical assistance, and the Vietnam Agricultural Research Institute (VASI). The Tam Dao Project, started in 1993, illustrates a development effort that takes many of the principles of integrated natural resources management (INRM) into consideration.

In identifying the problems faced by the community, both the livelihood concerns of the agrarian community as well as ecological linkages of the toposequence, from sloping land to valley bottom, are recognized and taken into consideration in seeking ecologically-sound land use types. For example one key intervention was the replacement of eucalyptus planting on the slopes, which exacerbated the soil erosion problem, with fruit trees intercropped with various annual crops that function as ground cover for erosion control, at the same time increasing farm income.

The project adopted a step-wise problem solving approach, emphasizing mutual learning and blending desirable elements of indigenous knowledge (e.g. innovative soil conservation practices for tree crops) with modern technological interventions (e.g. use of improved crop varieties and livestock breeds; livestock disease control and prevention measures). The agricultural production systems adopted also demonstrate good understanding of the linkages and flows of products between crops and livestock systems, and utilize these linkages effectively. A clear identification was also made of which interventions individual farm households can implement, and which require efforts at higher organizational levels (e.g. community-based seed production units, and deepening and maintenance of water reservoirs). Through testing of various combinations of crop and livestock systems and fitting these to the resource endowments and household objectives of the farmers, the Project has accumulated a rich knowledge base of plausible and promising innovations that are ecologically-sensitive as well as economically attractive to the local communities.

Appendix 2

The application of Multi-Agent Systems for addressing multi-scale INRM issues

Multi-Agent Systems (MAS), which is based on recent developments in the field of distributed artificial intelligence and cellular automata technology (Ferber, 1999), provides a means of representing knowledge and interactions among agents. In a MAS model, an agent is a computerized autonomous entity that is able to act locally in response to stimuli from the environment or in communication with other agents by sending messages and forming images of the world the way they perceive it (Bousquet et al. 1999). MAS models construct such agents and put them in realistic circumstances of interaction to look for answers to questions such as the coordination of players exploiting a given resource. Presently MAS is implemented on the Common-pool Resources and Multi-Agent Systems (Cormas) platform, which is a multi-agent simulation toolkit, running under the *VisualWorks* software, and specially designed for renewable resource management (Bousquet et al. 1998).

For the past ten years, MAS applications have been developed in the field of natural resources and environmental management (Bousquet, 1994; Le Page and Cury, 1996; Barreteau and Bousquet, 1998). Applied to INRM research, it focuses on the interactions between agro-ecological dynamics (i.e. the status and dynamics of the resources of concern) and socio-economic changes, basing on the hypothesis that the system dynamic is dependent on interactions between the resources and uses by both individual and collective agents (in this case, users) having different management strategies.

The design of the MAS model spatial support rests on spatial entities (e.g. cropped fields, forests), which are themselves another category of agents. When these entities yield resources, they are able to arbitrate their allocation between concurrent demands formulated by other agents (i.e. the users) exploiting these resources according to pre-defined rules or subjected to known constraints. The way agents exploit the resources of the spatial entities may depend on their own incomplete or partial representation of the environment. MAS allows for the construction of generic models to mitigate the site specificity of more classical type of INRM research. MAS can manipulate and incorporate into the same application or model spatial entities defined at different hierarchical levels, and can simulate behaviour or changes with time. Therefore it is possible to take into account the spatial hierarchy as well as the temporal dimension in representing and simulating system dynamics. Closely articulated with fieldwork, MAS is employed to provide a companion-modeling tool for researchers to work with diverse stakeholders.

An example is described below to illustrate how MAS simulation is applied to the soil erosion case study in North Thailand.

The spatial environment and land resources of the study area are depicted by GIS maps of topography and time series, plot-wise land use, which are transferred into the MAS environment.

Two inter-related spatial entities are considered in the model: homogeneous zones (with homogeneous slopes) and farmers' fields, with corresponding attributes including area, owner, slope characteristics (orientation, angle, length), crop being grown.

Two different classes of agents are included in the model, i.e. the communicating agents, passive situated agents. Two main categories of communicating agents are included:

1. The farming systems present in the area; there are three main types with attributes such as the amount of (land) resources (quantity and quality) and strategies regarding crop combinations.
2. The village, regulating the beginning and end of the crop year, farmers' actions in their fields, and pooling the results of the daily assessment of erosion risk in each field along the rainy season.

The passive situated agents are the crops in farmers' fields. Their attributes are crop calendars (early-late sowing dates, duration of the crop cycle), duration of their respective periods of susceptibility to erosion, sequences of cultivation practices, minimum and maximum size of their fields.

The model simulates the dynamics of land use changes. Young farmers inherit from old ones, and the farm type can change at the creation of new farms depending on the amount of resources they yield. Results of cash cropping activities in one year influence the crop allocation to the fields the next year, as well as off-farm activities in case of negative results/indebtness.

The control of the simulation can be set up according to a daily or yearly time scale. Actual long chronological series of daily rainfall data are used by the MAS model to assess soil erosion risk. The simulation results are depicted in land use change and corresponding erosion risk maps, as shown in Figure 5 of the main text.